

Asia Alrashed

AI / Machine Learning Engineer | Data Scientist | LLM & NLP

 UAE - Ajman

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Professional Summary

AI and Machine Learning Engineer with hands-on experience building end-to-end machine learning and data science solutions, from data preparation and feature engineering to model training, evaluation, deployment, and monitoring. Experienced in predictive modeling, NLP, and Large Language Models (LLMs), including transformer fine-tuning with Hugging Face and Parameter-Efficient Fine-Tuning (PEFT) techniques such as LoRA and QLoRA. Strong Python and SQL background with practical experience using PyTorch, TensorFlow, Scikit-learn, FastAPI, Docker, MLflow, PostgreSQL, RAG, and production-ready ML pipelines.

Professional Experience

AI Engineer | Data Scientist at Ocean Gate Line Company, UAE

Nov 2025 – Present

- Built an end-to-end machine learning and analytics workflow for logistics operations, covering data ingestion, cleaning, feature engineering, model development, evaluation, deployment, and business reporting.
- Designed automated ETL pipelines using Apache Airflow, Spark, Python, and FastAPI to process and validate three years of operational logistics data from multiple sources.
- Prepared and integrated customer, shipment, driver, collection, accounting, and delivery datasets using Pandas and SQL, resolving missing values, duplicates, inconsistent identifiers, and cross-source data relationship issues.
- Developed and evaluated an ETA prediction model using historical shipment data and engineered features to estimate delivery times and support proactive customer communication.
- Designed a relational data model and migrated cleaned multi-source datasets into PostgreSQL, creating reliable, structured datasets for machine learning and analytics workloads.
- Built and deployed LLM-powered conversational AI applications using LangChain, RAG pipelines, FastAPI, and vector-based retrieval, with a focus on Arabic NLP and Iraqi dialect adaptation. Prepared and refined conversational datasets to improve response naturalness, tone consistency, intent coverage, and dialect-specific customer engagement.
- Developed analytical and marketing attribution pipelines connecting customer interactions with advertising campaigns, improving campaign performance visibility and helping reduce underutilized advertising spend by 10–15%.

AI Engineer | ML | Data Scientist, Freelancer at SHAI FOR AI

Feb 2024 – Aug 2025

- Built end-to-end machine learning and deep learning solutions covering data preparation, feature engineering, model training, fine-tuning, evaluation, deployment, and reporting.
- Developed NLP and transformer-based solutions using Hugging Face Transformers, BERT, DistilBERT, PyTorch, and TensorFlow for text classification, sentiment analysis, semantic search, and other NLP tasks.
- Fine-tuned transformer models using Hugging Face Trainer and Datasets, managing tokenization, training configuration, evaluation, hyperparameter tuning, and model optimization.
- Applied Parameter-Efficient Fine-Tuning (PEFT) techniques, including LoRA and QLoRA, to efficiently adapt pre-trained language models while reducing trainable parameters and computational requirements.
- Developed LLM-based applications using LangChain, RAG, embeddings, vector retrieval, and prompt engineering for retrieval and conversational AI use cases.

- Trained and evaluated computer vision and deep learning models for image classification, medical imaging, segmentation, and multimodal applications.
- Deployed machine learning models and AI APIs using FastAPI and Docker, with experiment tracking and model lifecycle management using MLflow.

Tech skills

- **Programming:** Python, SQL, JavaScript, Java, C++
- **Data Science & Analysis:** Pandas, NumPy, EDA, Data Cleaning, Data Preprocessing, Feature Engineering, Data Validation, Statistical Analysis
- **Machine Learning & Deep Learning:** Scikit-learn, PyTorch, TensorFlow, Predictive Modeling, Classification, Regression, Clustering, Feature Engineering, Model Evaluation, Hyperparameter Tuning
- **LLMs, NLP & Fine-Tuning:** Hugging Face Transformers, BERT, DistilBERT, Parameter-Efficient Fine-Tuning (PEFT), LoRA, QLoRA, Hugging Face Trainer, Datasets, LangChain, RAG, Prompt Engineering, Embeddings, Semantic Search, Vector Retrieval
- **Data Engineering & Big Data:** ETL/ELT, Data Pipelines, Apache Spark, Apache Airflow, Hadoop, API-based Data Ingestion, Multi-source Data Integration, Data Migration, Data Reconciliation
- **MLOps & Deployment:** FastAPI, Docker, Kubernetes, MLflow, CI/CD, Model Deployment, Model Monitoring, API Development , AWS, Microsoft Azure
- **Cloud & Databases:** AWS EC2, AWS S3, PostgreSQL, MongoDB, MySQL .
- **Visualization & Reporting:** Matplotlib, Seaborn, Plotly, Power BI, Dashboarding, KPI Reporting, Business Reporting

Education

Damascuse University Faculty of IT Engineer

10 / 2016 - 11 / 2022

- Bachelor's Information Technology Engineer with a specialization in Artificial Intelligence

Highlighted Projects

- **LLM-Powered Conversational AI & Fine-Tuning**
Developed and deployed an LLM-powered conversational AI chatbot for customer and operational support. Fine-tuned language models using Hugging Face Transformers, Trainer, and Datasets, applying Parameter-Efficient Fine-Tuning (PEFT) techniques including LoRA and QLoRA for efficient model adaptation. Integrated the fine-tuned model with a Retrieval-Augmented Generation (RAG) pipeline using embeddings, vector retrieval, LangChain, and FastAPI to deliver context-aware responses based on company knowledge and support operational workflows.
- **Logistics Data Integration & ETA Prediction Pipeline**
Built an end-to-end FastAPI and Pandas-based ML pipeline to extract, clean, integrate, and migrate three years of logistics data from third-party web APIs into PostgreSQL. Preserved relationships across customers, orders, shipments, drivers, collections, and accounting records, then trained an ETA prediction ML model using historical data to estimate shipment arrival times and support proactive customer notifications.
- **Medical Imaging AI System**
Deep learning models for X-ray and 3D CBCT medical image analysis and classification.
- **Stock Market Sentiment Analysis System**
NLP pipeline using DistilBERT for financial sentiment analysis integrated into a stock ranking workflow.

Languages

- Arabic : Native
- English : Intermediate